화학특강 (2009년 2학기)
제1장

황현석

Units

1. Units
   1) SI prefix (metric prefix) (SI = Systemè International)

SI prefix	Symbol	Power of 10	SI prefix	Symbol	Power of 10
peta	P	10^{15}	milli	m	10^{-3}
tera	T	10^{12}	micro	μ	10^{-6}
giga	G	10^{9}	nano	n	10^{-9}
mega	M	10^{6}	pico	p	10^{-12}
kilo	k	10^{3}	femto	f	10^{-15}
centi	c	10^{-2}	atto	a	10^{-18}

Units

1. Units

   1) Volume

      - 1 mL = 1 cm^3 = 1 cc
      - 1 gallon = 3.785 L = 8 pint

   2) Mass and weight

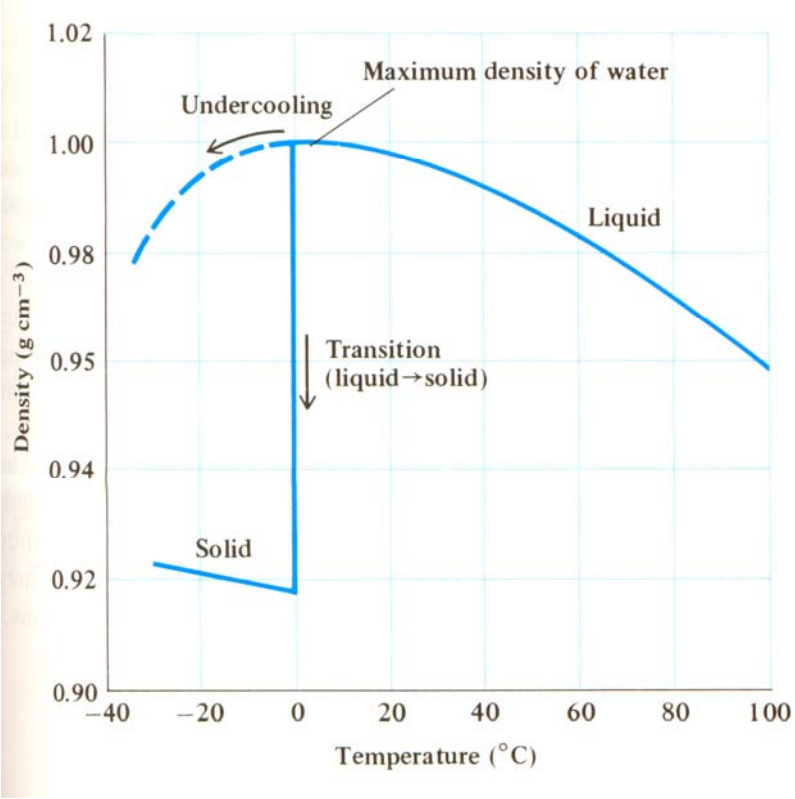
      - 1 lb ≈ 0.453 kg

   3) Density

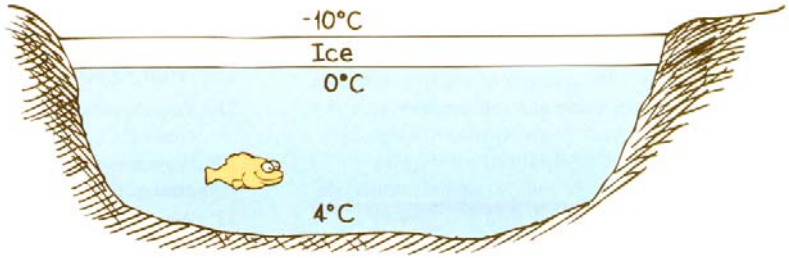
      - Density = Mass / Volume
      - Water density = 0.99707 g/cm^3 (25°C)
      - 물 1 L의 몰수?

Water density

- Water density



- Why do lakes freeze from the top down?

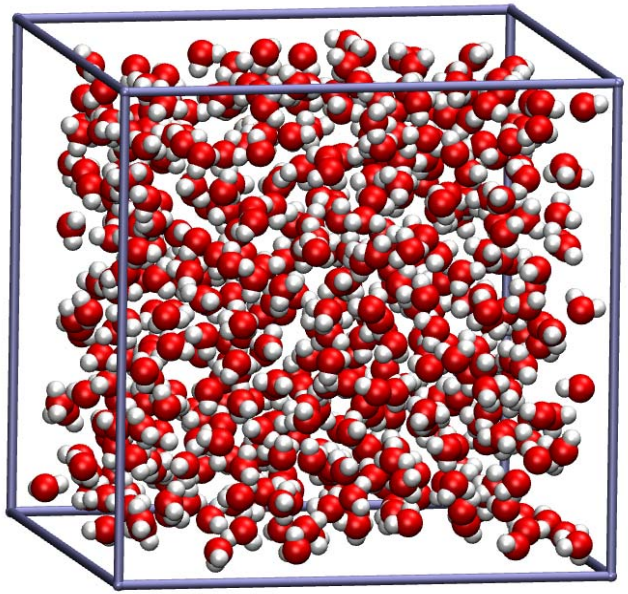


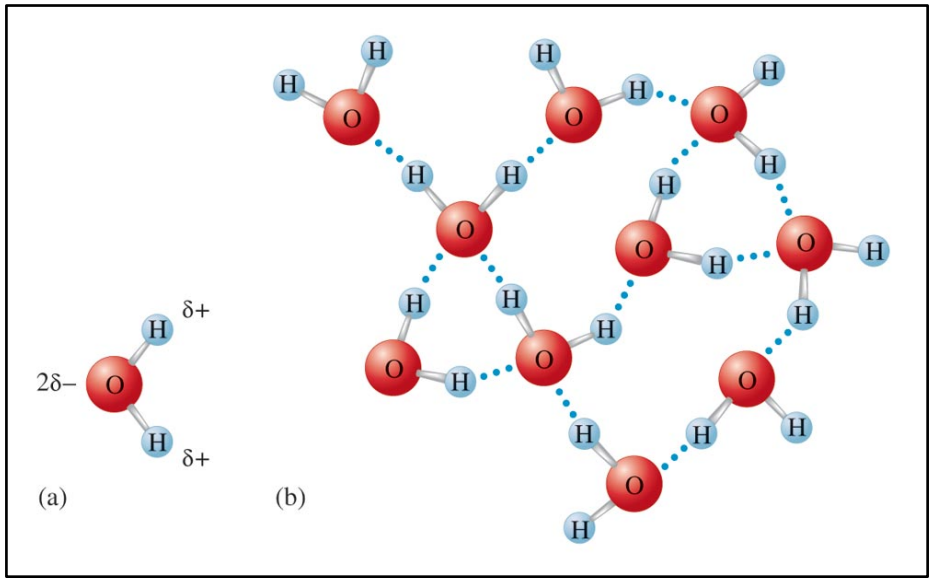
Thermodynamics of the Process: $H_2O(s) \rightarrow H_2O(l)$

T (°C)	T (K)	$\Delta H°$ (J/mol)	$\Delta S°$ (J/K · mol)	$\Delta G° = \Delta H° - T\Delta S°$ (J/mol)
−10	263	6.03×10^3	22.1	$+2.2 \times 10^2$
0	273	6.03×10^3	22.1	0
10	283	6.03×10^3	22.1	-2.2×10^2

물의 융해 과정에서 ΔH와 ΔS가 0보다 큰 이유는?

Structure of Liquid and Iced Water

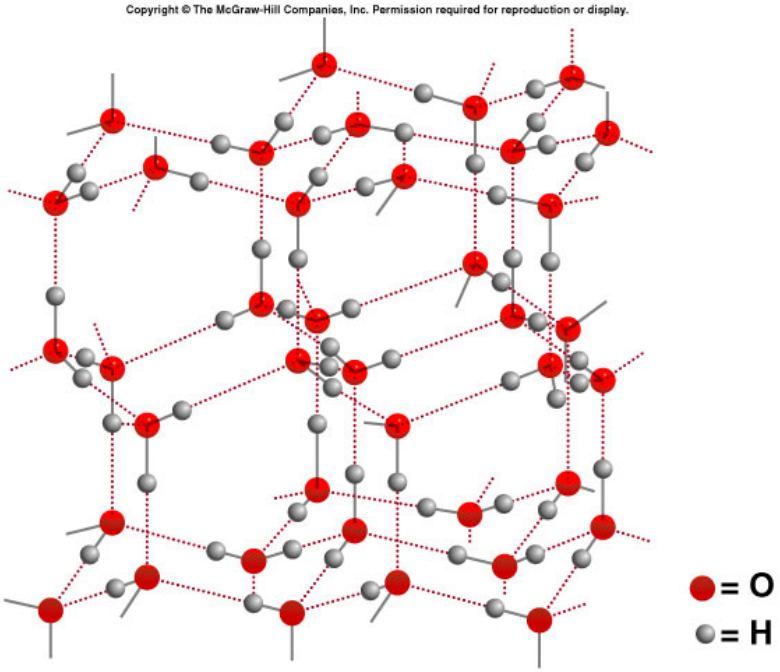




액체 상태에서 물분자들의 배열이 얼음 상태에서 물분자들의 배열보다 더 무질서하므로 액체 상태 물의 엔트로피가 더 크다. 점선은 산소원자와 수소원자 간의 수소결합을 나타낸다.

Structure of Iced Water (I)

- Three-dimensional structure of ice

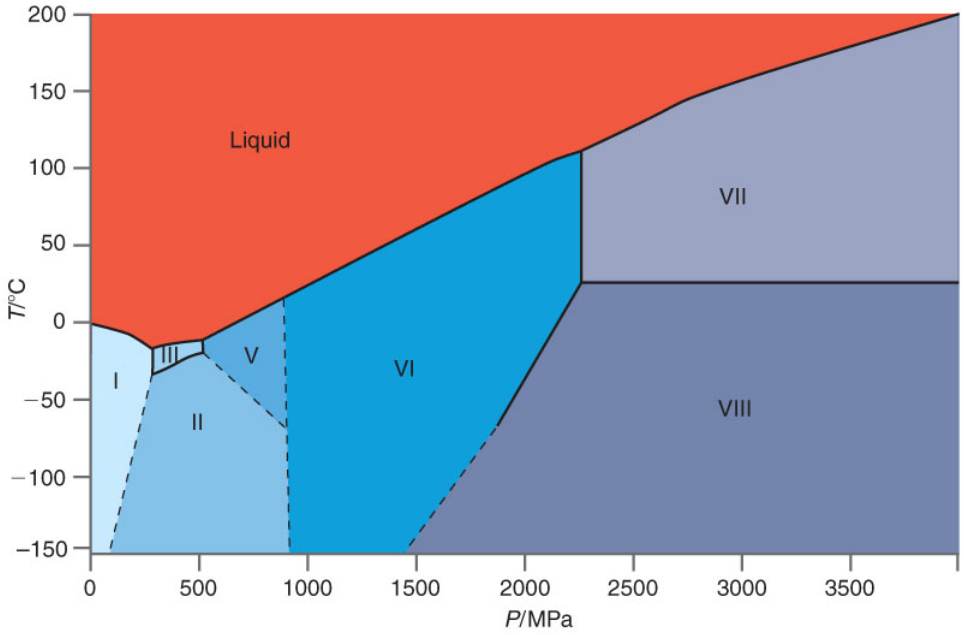


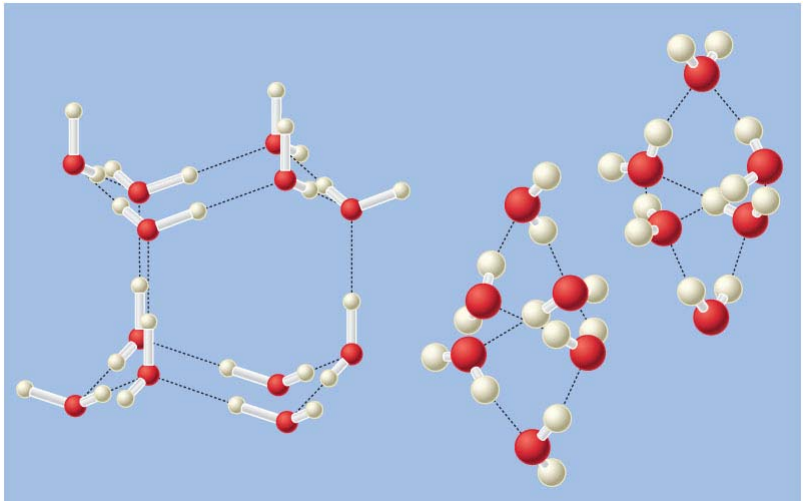


얼음의 3 차원 입체 구조. 한 개의 산소 원자는 네 개의 수소 원자와 공유 결합 혹은 수소 결합에 의해 연결되어 있다. 점선은 산소 원자와 수소 원자 간의 수소 결합을 나타낸다.

Structure of Iced Water (II)

- 온도와 압력에 따른 물의 서로 다른 고체상: 얼음의 여러가지 구조

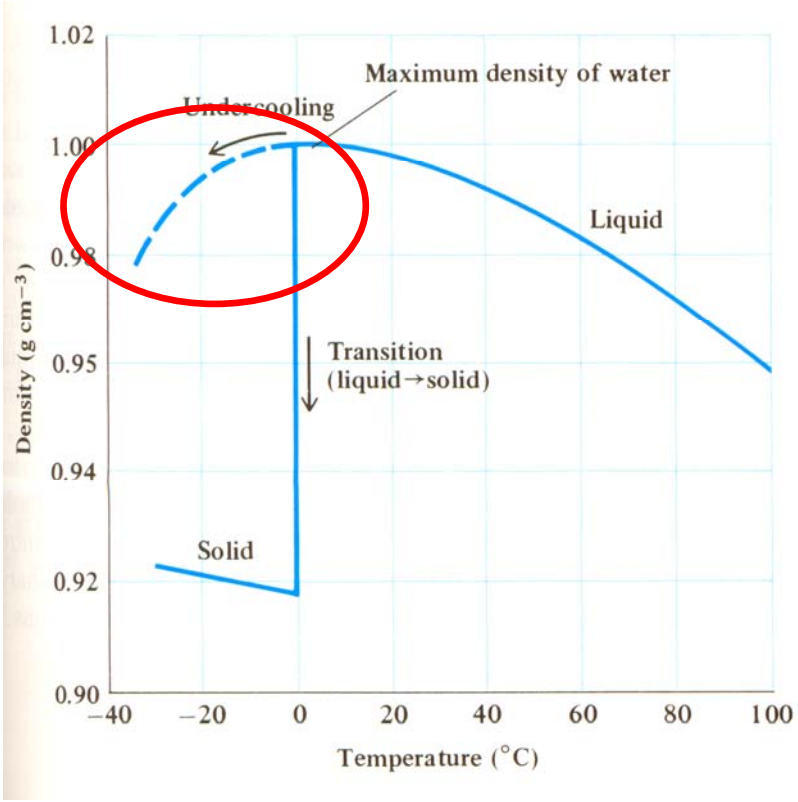




물의 서로 다른 두 가지 결정구조: 얼음 I과 얼음 VI

육방 얼음 (얼음 I)의 밀도는 0°C, 1기압에서 0.9167 g/cm^{3}이고 얼음 VI의 밀도는 1.6 GPa에서 1.31 g/cm^{3}이다. 이 압력에서 액체물의 밀도는 1.18 g/cm^{3}이다. 얼음 VI이 물에 뜰까? 가라앉을까?

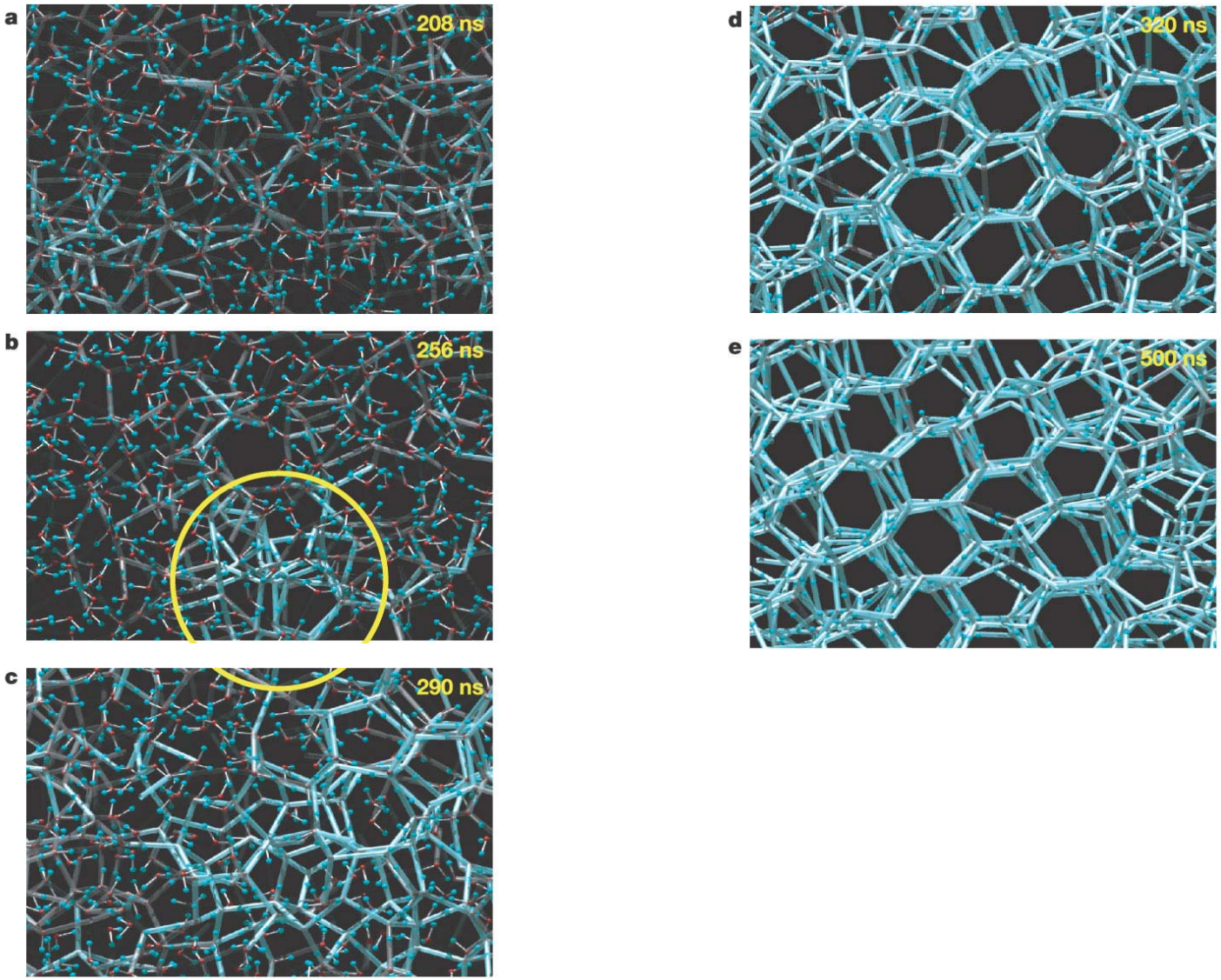
Supercooled Water



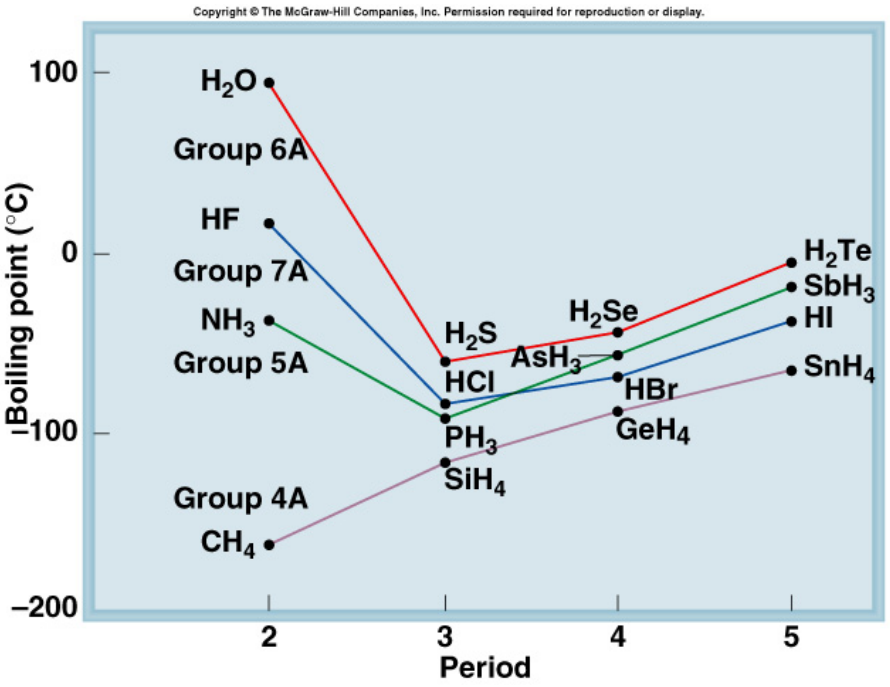


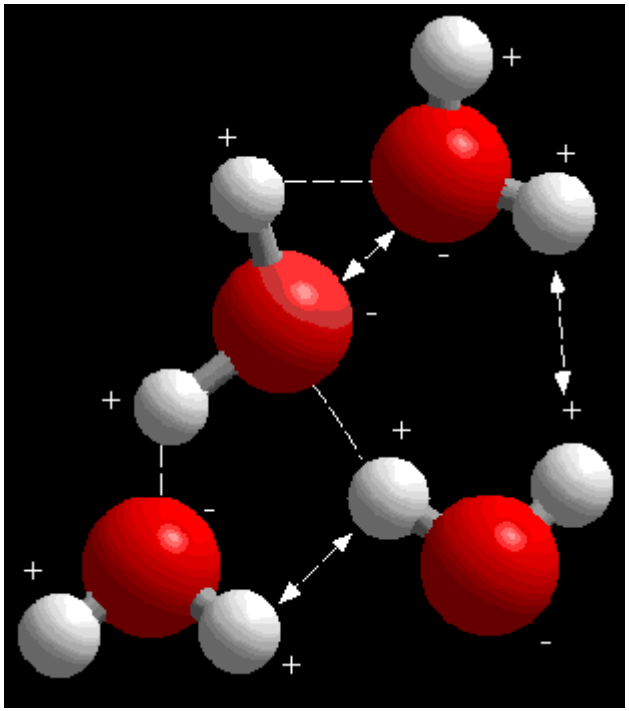


Supercooled Water: Molecular Dynamics Simulation (분자동역학 모의실험)



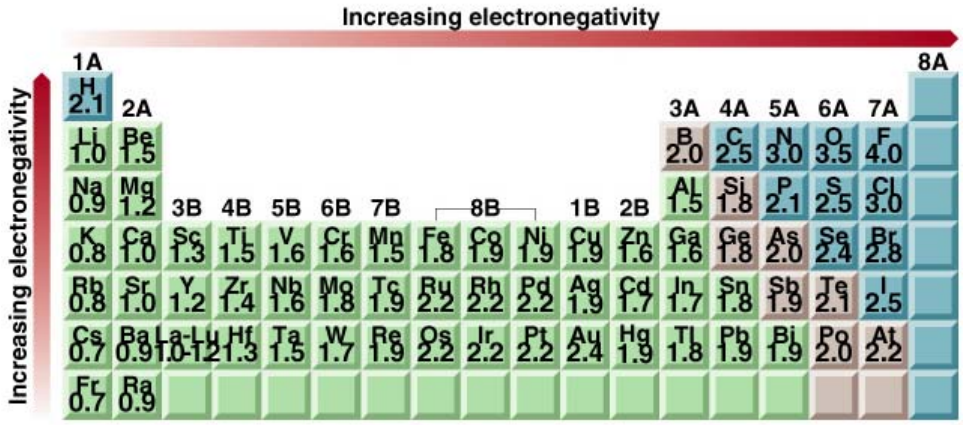
The Hydrogen Bond





Origin of the Hydrogen Bond

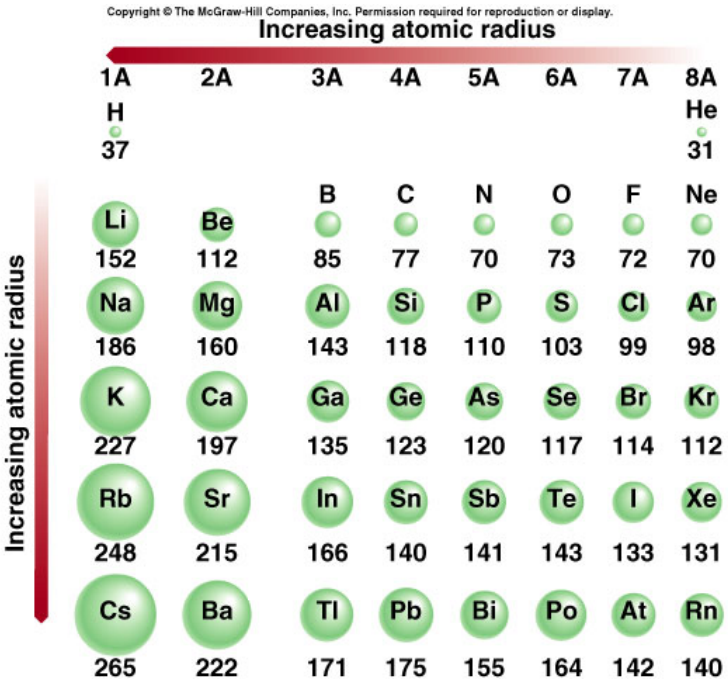
1. Large electronegativity difference between H and F, O, and N.
   - Electronegativity: the ability of an atom to attract toward itself the electrons in a chemical bond.
   - Linus Pauling



$$\chi_A - \chi_B = (eV)^{-1/2}\sqrt{E_d(AB) - [E_d(AA) + E_d(BB)]/2}$$

HBr 0.73 (dissociation energies:
H–Br, 3.79 eV; H–H, 4.52 eV; Br–Br 2.00 eV)

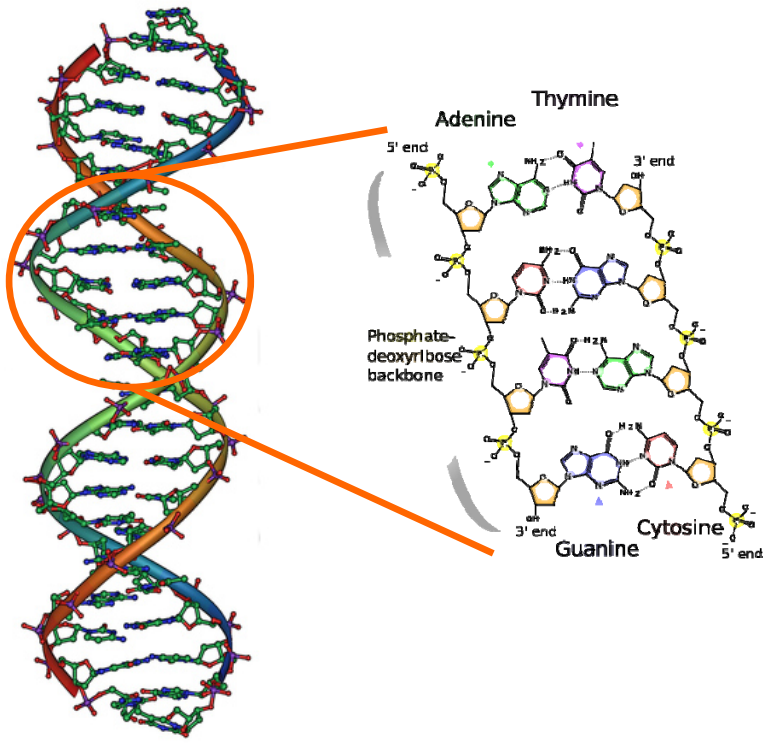
2. Small size of F, O, and N.



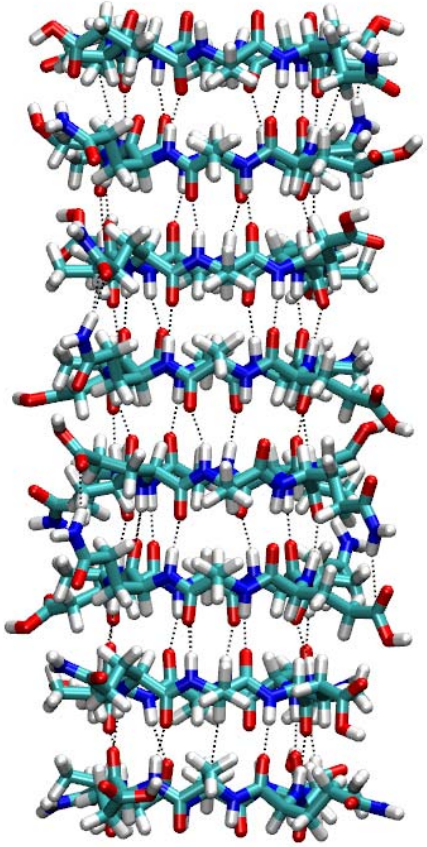
Chang, 8.3, 9.5, 9.10, and 11.2

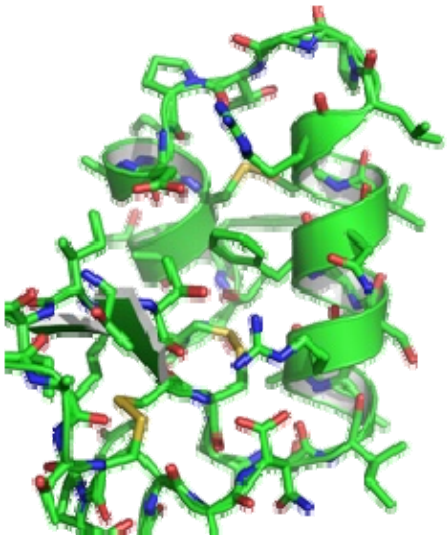
The Hydrogen Bond in Biological Systems

- DNA double helix

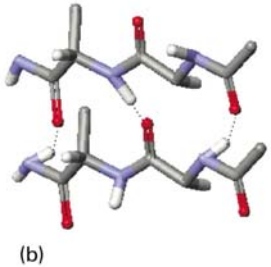


- Proteins









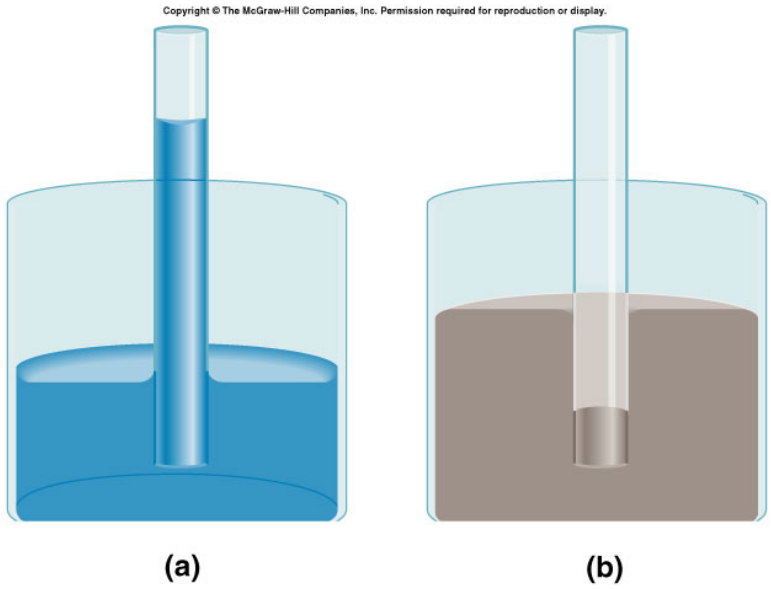
Surface Tension and Water Hydrogen Bond

- Surface tension





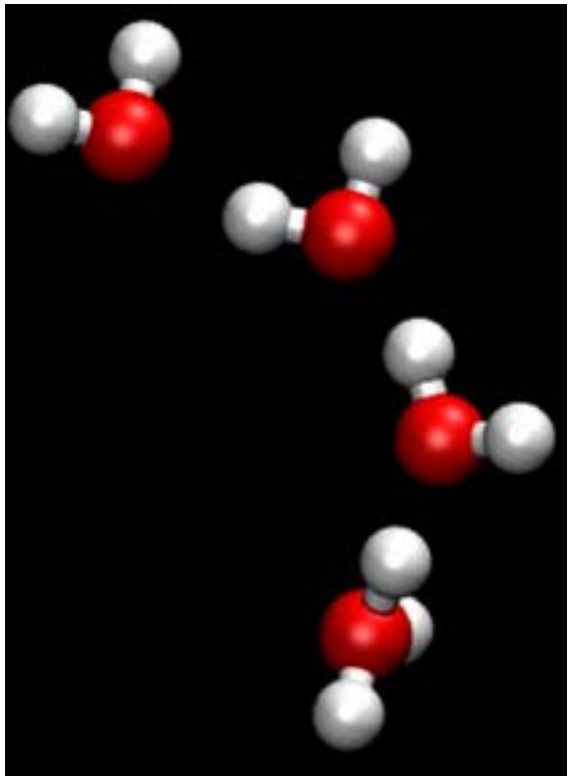
- Cohesion and adhesion
  - Adhesion: an attraction between unlike molecules.
  - Cohesion: the intermolecular attraction between like molecules.



Chang, 11.3

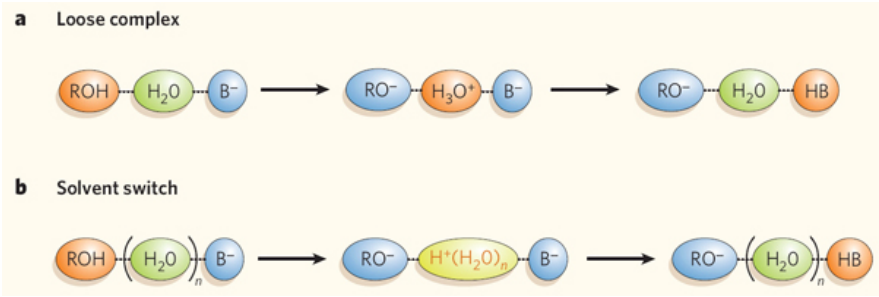
Grotthus Mechanism and Proton Transfer

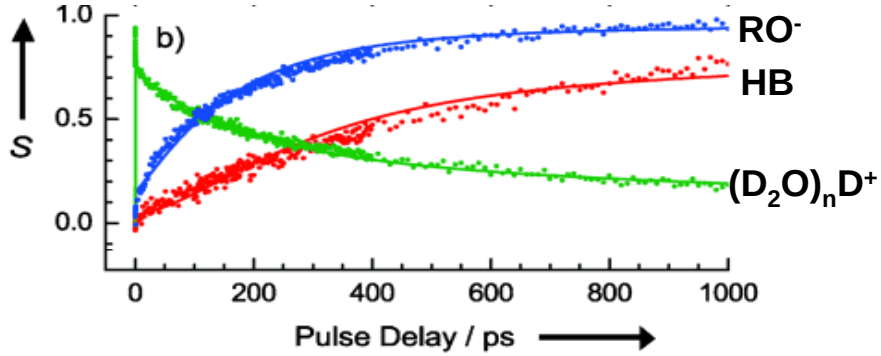
- Grotthus mechanism



Marx, ChemPhysChem **7**, 1849 (2006)

- Proton transfer in acid and base
  - Arrhenius definition
  - Brønsted definition
  - Lewis defintion



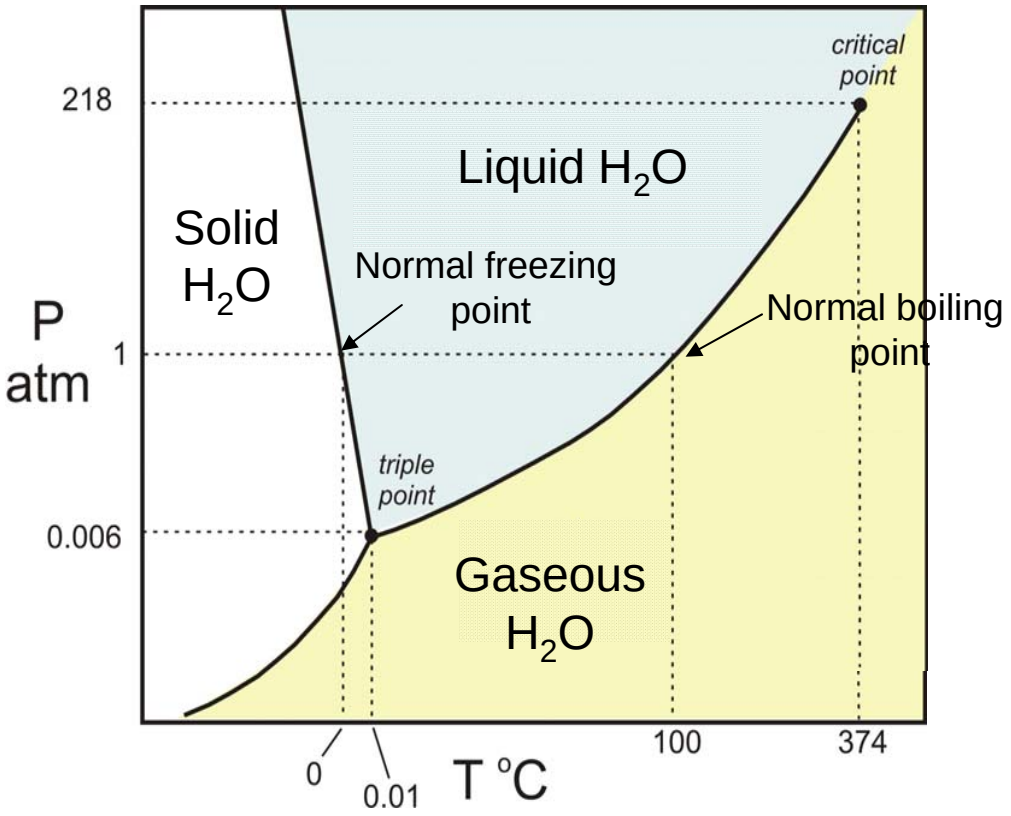


Mohammed et al., Angew. Chem. Int. Ed. **46**, 1458 (2007)

Chang, 15.1

상평형도 (Phase diagram)

- Water phase diagram



- 삼중점 (triple point): 세 상태가 서로 평형을 유지하는 유일한 상태.
- 임계온도 (critical temperature, T_c): 이 온도 이상에서는 아무리 높은 압력을 가하더라도 기체가 액화될 수 없는 온도.
- 임계압력 (critical pressure, P_c): 임계온도에서 액화시키기 위해 가해주는 최소 압력
- 임계점 (critical point): 상평형도에서 임계온도와 임계압력에 의해 정의되는 상태.
- 초임계유체 (supercritical fluid): 임계점을 지나 존재하는 물질의 상태.

Chang, General Chemistry 8th ed., Chapter 11

화학퍼텐셜

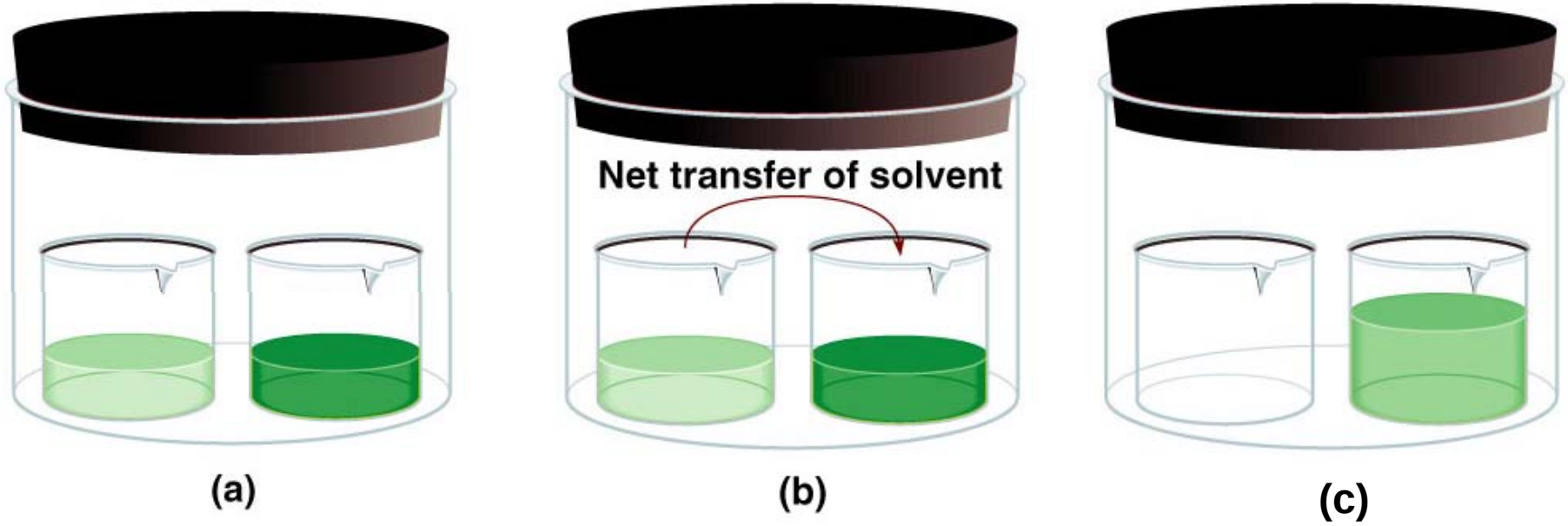
- 화학퍼텐셜: 일정한 상에 있는 물질이 다른 상으로 움직이려는 경향.
- 물질은 화학퍼텐셜이 높은 상에서 낮은 상으로 움직이려는 경향이 있다.
- 이상기체의 화학퍼텐셜

$$\mu(T,P) = \mu^{\circ}(T) + RT\ln\left(\frac{P}{P^{\circ}}\right)$$

- 이상액체의 화학퍼텐셜

$$\mu_{\text{용매}}(T,P) = \mu^{*}_{\text{용매}}(T,P) + RT\ln x_{\text{용매}}$$

Chemical Potential: Example



Clapeyron and Clasius-Clapeyron Equations

- Clapeyron equation

$$\frac{dP}{dT} = \frac{\Delta_{\text{fus}} H}{T \Delta_{\text{fus}} V}$$

$$P \approx P^* + \frac{\Delta_{\text{fus}} H}{\Delta_{\text{fus}} V}(T - T^*)$$

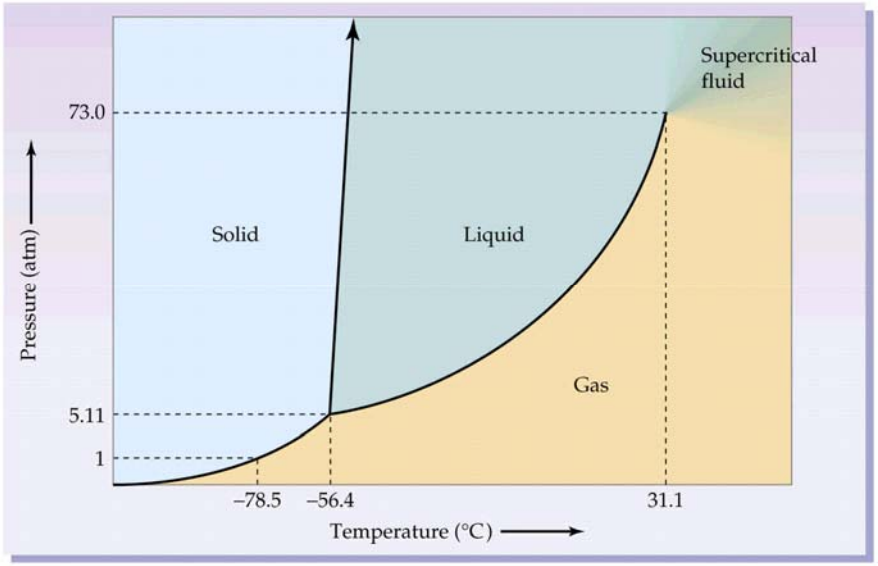
- Clausius-Clapeyron equation

$$\frac{dP}{dT} = \frac{\Delta_{\text{vap}} H}{T \Delta_{\text{vap}} V}$$

$$\ln\left(\frac{P}{P^*}\right) \approx -\frac{\Delta_{\text{fus}} H}{R}\left(\frac{1}{T} - \frac{1}{T^*}\right)$$

CO_2 Phase Diagram

- CO_2 Phase Diagram



- CO_2 Sublimation

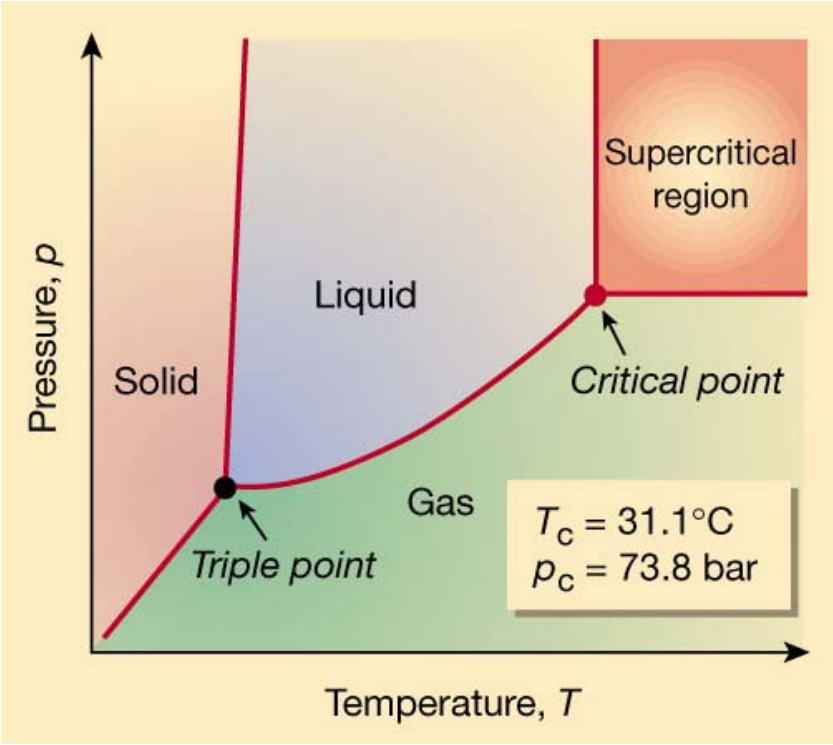


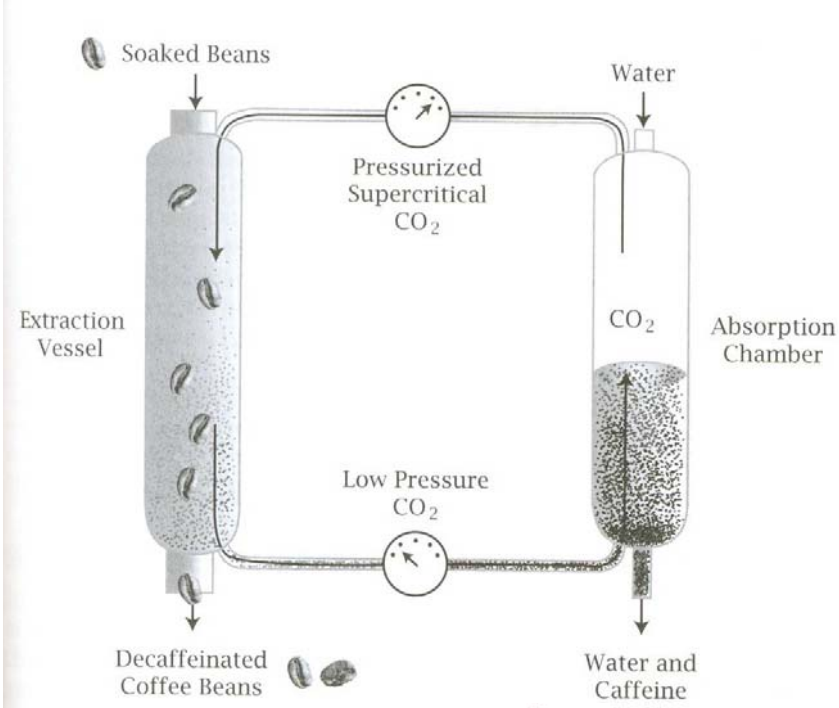
- CO_2 fire extinguisher



Supercritical CO_2

- Supercritical CO_2 ($scCO_2$)
- Decaffeination using $scCO_2$





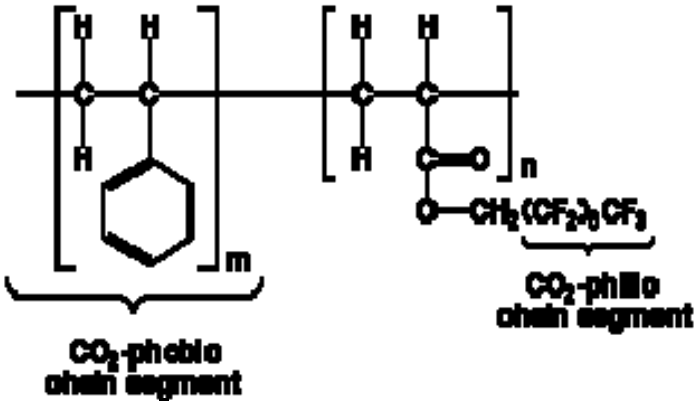


Supercritical CO_2

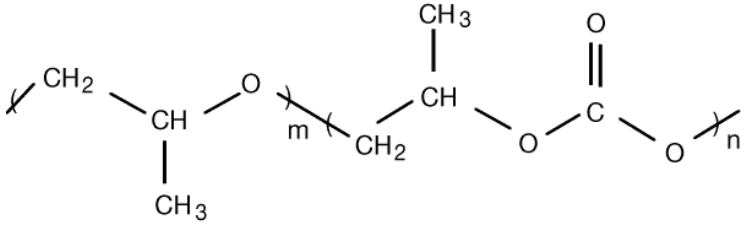
- **Advantages of scCO_2**
  - Ecological and economical benefits → no noxious and toxic residues once the solvent has been allowed to evaporate. (green chemistry)
  - A Unique combination of properties from both liquid and gas → low viscosity and high diffusion coefficients of solutes with a density like that of a liquide
- **Problems in scCO_2**
  - Poor solubility of many interesting target substances → Use of solubilizers (or stabilizer)
  - Polysiloxanes and fluorocarbons as solubilizers are expensive.
- **poly(ether_carbonate) copolymers as a solubilizer**
  - Inexpensive.
  - A favorable interaction of the carbonyl group with CO_2 may increase the enthalpy of mixing.

Atkins, Physical Chemistry 8th ed. pp119-120

- **Fluorocarbon copolymer**



- **poly(ether_carbonate) copolymers**



Sarbu et al., Nature 405, 165 (2000)